

# The Cell-Level Precision Problem in the AI Validation Design

AI Productivity Project: Current Writeup

August 4, 2026

## Purpose and source convention

This note asks whether the proposal can estimate the true productivity gain  $\theta_j$  precisely for a cell that does not receive an RCT. It first shows why the held-out-cell MSE contains a nonvanishing term  $\omega^2$ . It then reports a reproducible simulation of the MSE as the number of validated cells grows. Finally, it states what must change if precise cell-level estimation, rather than calibrated prediction, is the objective. It also isolates the exact point at which the proposal’s interpretation goes beyond its formal result.

The project source is *The Distributional Effects of AI’s Productivity Gains: A Data Fusion Approach*, especially pp. 2–4. Statements marked *Source* come from that proposal. Statements marked *Proposal claim* are asserted there without proof. Statements marked *Assumption* or *Calibration* are added here. Statements marked *Derived* follow from displayed assumptions. Statements marked *Simulation result* come from the program `ai_mse_simulation_2026-08-04.py`, archived at commit 4307f84.

## 1 What is observed and what remains unknown

### Notation

| Symbol                            | Meaning                                                                                                   |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------|
| $j = 1, \dots, J$                 | Occupation-task cell and total number of cells.                                                           |
| $R_j, m$                          | Validation indicator and number of validated cells, $m = \sum_j R_j$ .                                    |
| $\theta_j$                        | True productivity gain in cell $j$ .                                                                      |
| $S_j$                             | Broad model score observed in every cell.                                                                 |
| $\hat{v}_j, \varepsilon_j, s_j^2$ | RCT estimate in a validated cell, its error, and its design variance.                                     |
| $\mu, \tau^2$                     | Cross-cell mean and variance of true gains.                                                               |
| $a, b$                            | Score intercept and score slope; $0 < b < 1$ represents compression.                                      |
| $\eta_j, \sigma_S^2$              | Score error and its variance.                                                                             |
| $\psi, \psi_0$                    | Generic and true shared parameter vectors, ordered as $(\mu, \tau^2, a, b, \sigma_S^2)$ .                 |
| $\mathcal{D}_m$                   | Data from the $m$ validated cells.                                                                        |
| $h(S; \psi)$                      | Conditional-mean rule evaluated at score $S$ and shared parameter $\psi$ .                                |
| $m_j$                             | Oracle score-only prediction $\mathbb{E}[\theta_j \mid S_j, \psi_0] = h(S_j; \psi_0)$ .                   |
| $u_j$                             | Cell-specific difference $\theta_j - m_j$ .                                                               |
| $\omega^2$                        | Score-only conditional variance $\text{Var}(\theta_j \mid S_j, \psi_0)$ , constant in the Gaussian model. |

| Symbol                                                | Meaning                                                                                      |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------|
| $\widehat{\psi}_m, \widehat{\theta}_{j,m}$            | Shared-parameter estimator and held-out-cell prediction $h(S_j; \widehat{\psi}_m)$ .         |
| $\mathcal{R}_m, \Delta_m$                             | Held-out-cell MSE and its excess above the oracle floor.                                     |
| $\Lambda$                                             | Constant in the expansion $\Delta_m = \Lambda/m + o(m^{-1})$ .                               |
| $o(m^{-1})$                                           | A term that is small relative to $m^{-1}$ .                                                  |
| $\sigma_\theta^2, \beta, \nu_j, \widehat{\theta}_j^v$ | Proposal aliases for $\tau^2, b, \eta_j, \widehat{v}_j$ , used only in the conversion below. |

*Source-to-project notation conversion.* The proposal uses  $\sigma_\theta^2, \beta, \nu_j, \widehat{\theta}_j^v$ , while the project notes use  $\tau^2, b, \eta_j, \widehat{v}_j$ . This note keeps the project notation. The symbols  $\mathcal{D}_m, u_j, \mathcal{R}_m$ , and  $\Delta_m$  are added because the proposal does not name these objects.

The data differ between validated and unvalidated cells:

| Cell      | Observed data        | Information supplied                                                                                 |
|-----------|----------------------|------------------------------------------------------------------------------------------------------|
| $R_j = 1$ | $S_j, \widehat{v}_j$ | The score and a noisy direct measurement of this cell's gain.                                        |
| $R_j = 0$ | $S_j$ only           | The score-to-truth relation learned from other cells, but no direct measurement of this cell's gain. |

The proposal uses approximately  $J = 1,000$  cells and an experimental anchor of roughly  $m = 40$  cells. The MSE curve in Figure 2 is evaluated on held-out cells. *Source: proposal, pp. 1–3.*

The model for one basket is

$$\theta_j \sim N(\mu, \tau^2), \quad S_j = a + b\theta_j + \eta_j, \quad \eta_j \sim N(0, \sigma_S^2), \quad \eta_j \perp\!\!\!\perp \theta_j. \quad (1)$$

For a validated cell,

$$\widehat{v}_j = \theta_j + \varepsilon_j, \quad \varepsilon_j \sim N(0, s_j^2), \quad \varepsilon_j \perp\!\!\!\perp (\theta_j, \eta_j). \quad (2)$$

The Gaussian validation equation is from the proposal. Independence of the score and validation errors is made explicit here because it is required for the posterior calculation below. *Source and additional assumption: proposal, p. 4, translated into project notation.*

Normal updating gives

$$m_j = h(S_j; \psi_0) \quad (3)$$

$$= \mu + \frac{b\tau^2}{b^2\tau^2 + \sigma_S^2}(S_j - a - b\mu), \quad (4)$$

$$\omega^2 = \text{Var}(\theta_j \mid S_j, \psi_0) \quad (5)$$

$$= \frac{\tau^2\sigma_S^2}{b^2\tau^2 + \sigma_S^2}. \quad (6)$$

*Derived from (1).*

Write

$$\theta_j = m_j + u_j, \quad \mathbb{E}[u_j \mid S_j, \psi_0] = 0, \quad \mathbb{E}[u_j^2 \mid S_j, \psi_0] = \omega^2. \quad (7)$$

The validation sample can learn the distribution of  $u_j$ . It does not reveal the realized  $u_j$  for a cell with  $R_j = 0$ . Even when  $\psi_0$  is known, the score equation contains two unknown realized values,  $\theta_j$  and  $\eta_j$ , for one observed  $S_j$ .

For the plug-in prediction  $\hat{\theta}_{j,m}$ , Appendix A proves the exact decomposition

$$\mathcal{R}_m = \mathbb{E}[(\hat{\theta}_{j,m} - \theta_j)^2] = \omega^2 + \underbrace{\mathbb{E}[(\hat{\theta}_{j,m} - m_j)^2]}_{\Delta_m}. \quad (8)$$

Under the regularity conditions in Appendix A,

$$\Delta_m = \frac{\Lambda}{m} + o(m^{-1}), \quad \mathcal{R}_m = \omega^2 + \frac{\Lambda}{m} + o(m^{-1}). \quad (9)$$

*Derived.* The proposal asserts the  $m^{-1}$  excess-risk rate but does not supply its assumptions or proof. *Proposal claim:* p. 4, Appendix A.3.

The restriction  $R_j = 0$  is essential. If  $j$  is validated and the shared parameters are known, combining  $S_j$  and  $\hat{v}_j$  gives

$$\text{Var}(\theta_j \mid S_j, \hat{v}_j, \psi_0) = \left( \frac{1}{\omega^2} + \frac{1}{s_j^2} \right)^{-1} = \frac{\omega^2 s_j^2}{\omega^2 + s_j^2}. \quad (10)$$

*Derived from (1)–(2).* The  $\omega^2$  floor studied below applies to the cells that the proposal asks the model to extrapolate to, not to a cell that receives its own precise RCT.

## 2 Simulation of MSE as the validation anchor grows

### Notation

| Symbol                                                                    | Meaning                                                                                                |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $j, J, m$                                                                 | Cell index, total cells, and number of validated cells.                                                |
| $\theta_j, S_j, \eta_j, \hat{v}_j, \varepsilon_j, s_j^2$                  | True gain, broad score, score error, RCT estimate, validation error, and validation-estimate variance. |
| $\mu, \tau^2, a, b, \sigma_S^2$                                           | The five shared Gaussian channel parameters.                                                           |
| $\psi_0, \hat{\psi}_m$                                                    | True and fitted shared parameter vectors.                                                              |
| $h, m_j, \hat{\theta}_{j,m}$                                              | Conditional-mean rule, oracle score-only prediction, and fitted held-out prediction.                   |
| $\omega^2, \mathcal{R}_m, \Delta_m, \Lambda$                              | Oracle floor, total held-out MSE, excess MSE, and its first-order coefficient.                         |
| $\rho$                                                                    | Score-truth correlation $\text{Corr}(S_j, \theta_j)$ .                                                 |
| $D_S, \mu_S$                                                              | True score variance $b^2 \tau^2 + \sigma_S^2$ and mean $a + b\mu$ .                                    |
| $c, \hat{c}_m$                                                            | True and fitted conditional-mean slopes, $c = b\tau^2/D_S$ .                                           |
| $N_{\text{MC}}$                                                           | Number of Monte Carlo repetitions.                                                                     |
| $r = 1, \dots, N_{\text{MC}}$                                             | Monte Carlo replication index.                                                                         |
| $\mathcal{M}$                                                             | Simulated validation-budget grid.                                                                      |
| $\bar{S}_m, \bar{v}_m$                                                    | Validation-sample means of the score and RCT estimate.                                                 |
| $\hat{V}_{S,m}, \hat{V}_{v,m}, \hat{C}_{Sv,m}$                            | Validation-sample score variance, RCT-estimate variance, and their covariance.                         |
| $[x]_+, \Pi_{[0,1]}(x)$                                                   | Numerical projections $\max(x, 10^{-10})$ and $\min\{1, \max(0, x)\}$ .                                |
| $\hat{\mu}_m, \hat{\tau}_m^2, \hat{a}_m, \hat{b}_m, \hat{\sigma}_{S,m}^2$ | Moment estimators of the five shared parameters.                                                       |
| $\hat{\Lambda}$                                                           | Coefficient fitted to the simulated relation $\Delta_m \simeq \hat{\Lambda}/m$ .                       |

| Symbol                                                                                | Meaning                                                                                                  |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| $\Delta_m^{(r)}, \widehat{\Delta}_m^{\text{MC}}, \widehat{\mathcal{R}}_m^{\text{MC}}$ | Integrated excess risk in replication $r$ , its Monte Carlo mean, and the corresponding total-risk mean. |
| MCSE                                                                                  | Monte Carlo standard error of a simulated mean.                                                          |

## Calibration

The proposal says that  $J$  is on the order of one thousand, uses approximately  $m = 40$  validated cells in its illustration, reports a score-truth correlation near 0.44, and illustrates compression with a slope near 0.4. *Source: proposal, pp. 1–3 and p. 5.* The value on p. 5 is a Spearman correlation. The Gaussian simulation sets the Pearson correlation  $\rho$  to the same numerical value as an illustrative calibration; the proposal does not imply that these two correlations are equal. *Calibration qualification.*

The primary simulation uses

$$\mu = 30, \quad \tau = 10, \quad b = 0.4, \quad \rho = 0.44, \quad a = (1 - b)\mu = 18, \quad s_j = 3. \quad (11)$$

Gains and standard deviations are in percentage points. The choices  $\mu = 30, \tau = 10$ , and  $s_j = 3$  are illustrative rather than empirical estimates. *Calibration.*

The score-noise variance is chosen to match  $\rho$ :

$$\rho = \frac{b\tau}{\sqrt{b^2\tau^2 + \sigma_S^2}}, \quad (12)$$

$$\sigma_S^2 = b^2\tau^2(\rho^{-2} - 1) = 66.64. \quad (13)$$

Hence  $\sigma_S = 8.16$ . The oracle floor is

$$\omega^2 = \tau^2(1 - \rho^2) = 80.64, \quad \sqrt{\omega^2} = 8.98. \quad (14)$$

*Derived from (6) and the calibration.*

## Method

The simulation uses  $J = 1,000$ , seed 20260804, and

$$\mathcal{M} = \{10, 20, 40, 80, 160, 320, 640\}. \quad (15)$$

The value  $m = 640$  is included only to display the rate farther along the curve; it is not part of the proposal's  $m \ll J$  benchmark. For each Monte Carlo repetition, the simulation performs the following steps. Hatted sample moments depend on  $r$ ; that superscript is suppressed inside the steps.

1. Draw  $J$  cells from (1) and draw validation estimates from (2) with common variance  $s_j^2 = 9$ . The first  $m$  cells form a nested validation anchor. Because the cells are exchangeable, this is distributionally equivalent to selecting  $m$  cells at random.

2. Estimate the shared channel by the proposal's identifying moments. The sample variances and covariance below use denominator  $m - 1$ :

$$\hat{\mu}_m = \bar{v}_m, \quad (16)$$

$$\hat{\tau}_m^2 = [\hat{V}_{v,m} - s_j^2]_+, \quad (17)$$

$$\hat{b}_m = \Pi_{[0,1]} \left( \frac{\hat{C}_{Sv,m}}{\hat{\tau}_m^2} \right), \quad (18)$$

$$\hat{a}_m = \bar{S}_m - \hat{b}_m \hat{\mu}_m, \quad (19)$$

$$\hat{\sigma}_{S,m}^2 = [\hat{V}_{S,m} - \hat{b}_m^2 \hat{\tau}_m^2]_+. \quad (20)$$

The projections enforce the parameter space only when a small validation sample produces invalid raw moments. They are used in 15.3 percent of repetitions at  $m = 10$ , 3.85 percent at  $m = 20$ , 0.4 percent at  $m = 40$ , and zero repetitions at  $m \geq 80$ .

3. Form  $\hat{\psi}_m$ , calculate  $\hat{c}_m = \hat{b}_m \hat{\tau}_m^2 / (\hat{b}_m^2 \hat{\tau}_m^2 + \hat{\sigma}_{S,m}^2)$ , and predict a held-out cell by

$$\hat{\theta}_{j,m} = \hat{\mu}_m + \hat{c}_m (S_j - \bar{S}_m). \quad (21)$$

4. Integrate over a new held-out score conditional on the fitted rule. The exact excess risk in repetition  $r$  is

$$\Delta_m^{(r)} = \{\hat{\mu}_m + \hat{c}_m (\mu_S - \bar{S}_m) - \mu\}^2 + (\hat{c}_m - c)^2 D_S. \quad (22)$$

The reported simulation means are

$$\hat{\Delta}_m^{\text{MC}} = \frac{1}{N_{\text{MC}}} \sum_{r=1}^{N_{\text{MC}}} \Delta_m^{(r)}, \quad \hat{\mathcal{R}}_m^{\text{MC}} = \omega^2 + \hat{\Delta}_m^{\text{MC}}. \quad (23)$$

This removes evaluation-sample noise but does not replace the simulated validation anchor. A separate calculation predicts the remaining  $J - m$  cells directly; its mean MSE differs from the integrated risk by at most 0.073 MSE units over the simulated grid.

5. Begin with  $N_{\text{MC}} = 2,000$  repetitions. Increase that number only if the largest relative MCSE for  $m \geq 40$  exceeds 3 percent. The run stopped at 2,000 because the largest relative MCSE was 2.37 percent. A run with 10,000 repetitions was therefore unnecessary for the stated accuracy target.

*Simulation method.* The program uses the code names `sigma_theta` and `beta` for the project symbols  $\tau$  and  $b$ . All results below are generated by the archived code; none are proposal estimates.

## Results

| $m$ | $\hat{\Delta}_m^{\text{MC}}$ | $\hat{\mathcal{R}}_m^{\text{MC}}$ | $m \hat{\Delta}_m^{\text{MC}}$ | $\omega^2 / \hat{\mathcal{R}}_m^{\text{MC}}$ |
|-----|------------------------------|-----------------------------------|--------------------------------|----------------------------------------------|
| 10  | 19.940                       | 100.580                           | 199.4                          | 80.2%                                        |
| 20  | 9.988                        | 90.628                            | 199.8                          | 89.0%                                        |
| 40  | 5.013                        | 85.653                            | 200.5                          | 94.1%                                        |
| 80  | 2.331                        | 82.971                            | 186.5                          | 97.2%                                        |
| 160 | 1.129                        | 81.769                            | 180.7                          | 98.6%                                        |
| 320 | 0.579                        | 81.219                            | 185.4                          | 99.3%                                        |
| 640 | 0.283                        | 80.923                            | 180.9                          | 99.7%                                        |

*Simulation result.* The first risk column estimates the shrinking shared-parameter term  $\Delta_m$ ; the second estimates  $\mathcal{R}_m = 80.64 + \Delta_m$ .

The fitted relation is

$$\mathcal{R}_m \simeq 80.64 + \frac{196.7}{m}. \quad (24)$$

Regressing  $\log \hat{\Delta}_m^{\text{MC}}$  on  $\log m$  for  $m \geq 40$  gives slope  $-1.031$ . The stability of  $m\Delta_m$  and the slope near  $-1$  support the proposed  $m^{-1}$  rate under this data-generating process. They do not prove the rate outside it. *Simulation result.*

#### Held-out MSE as the randomized validation anchor grows

Proposal calibration:  $\text{corr}(\text{score}, \text{truth}) = 0.44$ ;  $\beta = 0.4$ ; validation SE = 3 pp

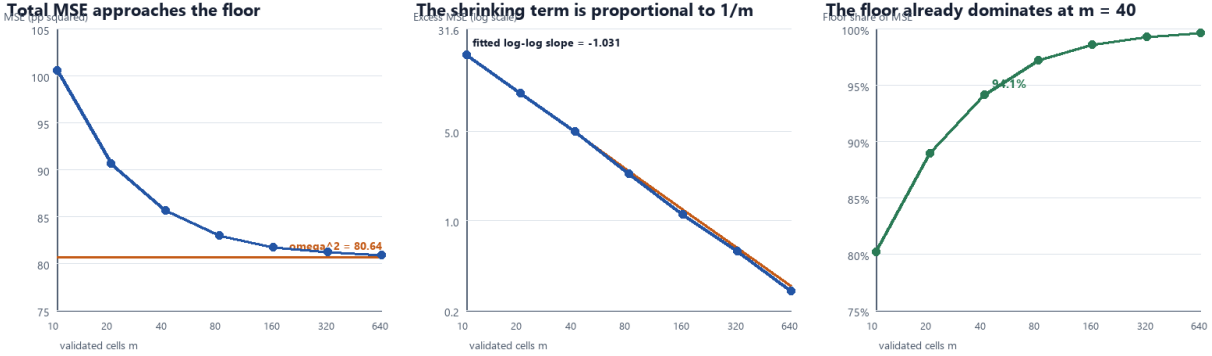


Figure 1: Held-out-cell MSE, its excess above the oracle floor, and the floor’s share of total MSE. The plot is generated by the archived simulation code with the calibration in (11).

At  $m = 40$ , the shrinking term is 5.013 while the floor is 80.64. Increasing  $m$  from 40 to infinity reduces RMSE only from  $\sqrt{85.653} = 9.25$  to  $\sqrt{80.64} = 8.98$  percentage points. Thus the proposal’s forty-cell benchmark is near the knee for learning the shared channel, but it does not imply precise prediction of an unvalidated cell.

More generally, the floor dominates when

$$\omega^2 \gg \frac{\Lambda}{m}, \quad \text{equivalently} \quad m \gg \frac{\Lambda}{\omega^2}. \quad (25)$$

Using  $\hat{\Lambda} = 196.7$ , the fitted  $\Lambda/m$  term is below 10 percent of the floor at approximately  $m = 25$ . *Derived from the fitted simulation relation.*

The conclusion is not that  $\omega^2$  dominates for every possible parameter vector. Its existence whenever  $\sigma_S^2 > 0$  is a feature of the model. Its numerical dominance depends on  $\omega^2$ ,  $\Lambda$ , and  $m$ . Within the simulated sensitivity grid, however, it is robust:

| $\rho$ | RCT SE | $\omega^2$ | $\Delta_{40}$ | Floor share |
|--------|--------|------------|---------------|-------------|
| 0.30   | 3      | 91.00      | 5.36          | 94.4%       |
| 0.44   | 3      | 80.64      | 5.01          | 94.1%       |
| 0.70   | 3      | 51.00      | 3.31          | 93.9%       |
| 0.44   | 1      | 80.64      | 4.58          | 94.6%       |
| 0.44   | 6      | 80.64      | 6.39          | 92.7%       |

*Simulation result.* Changing RCT precision changes the shrinking term, not the score-only floor. Changing score quality changes the floor itself. The table establishes robustness over these five choices, not over all possible data-generating processes.

### 3 What dominance implies for risk measurement and for $\theta_j$

#### Notation

| Symbol                                                 | Meaning                                                                                                |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $j, m$                                                 | Held-out cell and number of validated cells.                                                           |
| $\theta_j, S_j, \mathcal{D}_m, \psi_0$                 | Held-out true gain, its score, validation data from other cells, and the true shared parameter vector. |
| $m_j, u_j, \omega^2$                                   | Oracle score-only mean, cell-specific residual, and its variance.                                      |
| $\hat{\theta}_{j,m}, \mathcal{R}_m, \Delta_m, \Lambda$ | Fitted prediction, total MSE, excess MSE, and its first-order coefficient.                             |
| $\hat{\mathcal{R}}_m, \hat{\omega}^2, \hat{\Delta}_m$  | Estimators of total risk, the floor, and their difference $\hat{\mathcal{R}}_m - \hat{\omega}^2$ .     |
| $K_R, K_\omega$                                        | Effective numbers of independent observations used to estimate total risk and the floor.               |
| $z_{.975}$                                             | 97.5th percentile of the standard normal distribution, approximately 1.96.                             |
| $T_j$                                                  | Any estimator of $\theta_j$ based on the held-out cell's score and the validation data.                |
| $O_p(\cdot), \longrightarrow_p$                        | Order in probability and convergence in probability.                                                   |

#### The shrinking term is a small difference between large objects

Suppose one tries to estimate the shrinking term by subtraction:

$$\hat{\Delta}_m = \hat{\mathcal{R}}_m - \hat{\omega}^2. \quad (26)$$

Then

$$\hat{\Delta}_m - \Delta_m = (\hat{\mathcal{R}}_m - \mathcal{R}_m) - (\hat{\omega}^2 - \omega^2). \quad (27)$$

Under ordinary square-root rates,

$$\hat{\mathcal{R}}_m - \mathcal{R}_m = O_p(K_R^{-1/2}), \quad \hat{\omega}^2 - \omega^2 = O_p(K_\omega^{-1/2}). \quad (28)$$

Because  $\Delta_m \asymp m^{-1}$ , relative consistency of (26) requires

$$K_R^{-1/2} = o(m^{-1}), \quad K_\omega^{-1/2} = o(m^{-1}), \quad (29)$$

or

$$K_R \gg m^2, \quad K_\omega \gg m^2. \quad (30)$$

*Derived.* If  $\omega^2$  is estimated from the same  $m$  validation cells, its ordinary error is  $O_p(m^{-1/2})$ , which is larger than the  $O(m^{-1})$  object being recovered. Direct subtraction therefore cannot isolate  $\Lambda/m$  at relative precision without additional structure or a cancellation-based estimator. Indeed, ignoring the total-risk error only makes the problem easier, yet

$$m(\hat{\Delta}_m - \Delta_m) = -m(\hat{\omega}^2 - \omega^2) = O_p(\sqrt{m}), \quad (31)$$

which grows rather than vanishes. *Derived under the ordinary square-root rate for  $\hat{\omega}^2$ .*

An optimistic Gaussian calculation makes the scale concrete. If  $\omega^2$  is known and  $K_R$  independent, mean-zero Gaussian prediction errors have variance  $\mathcal{R}_m$ , then the standard error of their mean

squared error is approximately  $\sqrt{2\mathcal{R}_m}/\sqrt{K_R}$ . Detecting  $\Delta_m$  at the 95 percent normal threshold requires

$$K_R \geq 2z_{.975}^2 \left( \frac{\mathcal{R}_m}{\Delta_m} \right)^2. \quad (32)$$

For the primary simulation this gives:

| $m$ | $\Delta_m$ | $\omega^2/\Delta_m$ | Required $K_R$ |
|-----|------------|---------------------|----------------|
| 40  | 5.013      | 16.1                | 2,243          |
| 80  | 2.331      | 34.6                | 9,732          |
| 160 | 1.129      | 71.4                | 40,290         |
| 320 | 0.579      | 139.2               | 150,980        |

*Derived from the Gaussian variance of a squared error and the simulated risks.* Dependence across cells or estimation of  $\omega^2$  would increase the requirement.

The simulation detected the rate with 2,000 repetitions because it knew the simulated  $\omega^2$  and used the integrated expression (22). It did not recover a small unknown difference by subtracting two noisy empirical MSE estimates. This distinction is necessary before carrying the result to real data.

### An unvalidated cell effect is not consistently estimable

The word *cell* is important. The proposal defines  $\theta_j$  as the productivity gain in occupation-task cell  $j$ . It is a cell-average causal effect, not an individual worker's treatment effect. The question here is whether the realized effect of one particular unvalidated cell can be estimated precisely. *Source: proposal, pp. 1–2.*

The stronger problem concerns  $\theta_j$  itself. From (7),

$$\hat{\theta}_{j,m} - \theta_j = (\hat{\theta}_{j,m} - m_j) - u_j. \quad (33)$$

Under the regularity conditions behind (9),

$$\hat{\theta}_{j,m} - m_j = O_p(m^{-1/2}), \quad (34)$$

so

$$\hat{\theta}_{j,m} - \theta_j = -u_j + O_p(m^{-1/2}). \quad (35)$$

If  $\omega^2 > 0$ , the right side does not converge to zero. Therefore

$$\hat{\theta}_{j,m} \not\rightarrow_p \theta_j, \quad \text{MSE}(\hat{\theta}_{j,m}) \rightarrow \omega^2. \quad (36)$$

*Derived.*

This is not a defect of the plug-in estimator. If  $T_j$  is any estimator measurable with respect to  $(S_j, \mathcal{D}_m)$ , conditional variance decomposition gives

$$\mathbb{E}[(T_j - \theta_j)^2] = \mathbb{E}[(T_j - \mathbb{E}[\theta_j | S_j, \mathcal{D}_m])^2] \quad (37)$$

$$+ \mathbb{E}[\text{Var}(\theta_j | S_j, \mathcal{D}_m)] \quad (38)$$

$$\geq \mathbb{E}[\text{Var}(\theta_j | S_j, \mathcal{D}_m)]. \quad (39)$$

As the shared parameter becomes known, the last term converges to  $\omega^2$ . The posterior mean attains the lower bound. No other estimator using the same information can remove it. *Derived from conditional variance decomposition and held-out-cell independence.*



Under the primary Gaussian calibration, the limiting posterior standard deviation is 8.98 percentage points. A score-only 95 percent posterior interval has approximate half-width

$$1.96\sqrt{\omega^2} = 17.6 \quad (40)$$

percentage points. The simulation therefore reflects the mathematical identification problem directly: after the  $1/m$  component becomes small, almost all remaining prediction error is uncertainty about the particular held-out cell.

### The exact identification mistake in the proposal

The proposal correctly distinguishes the shared score-to-truth channel from the cell effects when it writes down the model. The mistake occurs when it passes from identifying the shared channel to claiming recovery of the realized map. Page 2 says that the validation pairs identify the common score-to-truth channel and that empirical Bayes will “recover the map.” Appendix A.2 is titled “The randomized anchor identifies everything.” The Figure 1 caption similarly says that the calibrated estimates “recover the real structure of who gains.” *Source: proposal, pp. 1–2 and p. 4.*

Write the shared parameter vector as

$$\psi_0 = (\mu, \tau^2, a, b, \sigma_S^2). \quad (41)$$

The randomized validation cells can consistently estimate this common vector under the proposal’s model:

$$\hat{\psi}_m \longrightarrow_p \psi_0. \quad (42)$$

The corresponding plug-in rule therefore satisfies

$$\hat{\theta}_{j,m} = h(S_j; \hat{\psi}_m) \quad (43)$$

$$\longrightarrow_p h(S_j; \psi_0) = \mathbb{E}[\theta_j \mid S_j, \psi_0] = m_j. \quad (44)$$

*Derived under M1–M4 in Appendix A.*

The invalid implication is

$$\hat{\psi}_m \longrightarrow_p \psi_0 \not\Rightarrow \hat{\theta}_{j,m} \longrightarrow_p \theta_j. \quad (45)$$

Identifying  $\psi_0$  identifies the conditional distribution of  $\theta_j$  given  $S_j$ , including its posterior mean. It does not reveal the realized residual  $u_j = \theta_j - m_j$ . Because  $\text{Var}(u_j \mid S_j, \psi_0) = \omega^2 > 0$ , different values of  $\theta_j$  remain compatible with the same observed score even after the common channel is known exactly. *Derived from (6)–(7).*

Appendix A.3 is not wrong when it states that the feasible estimator’s *excess risk over the oracle* is  $\Lambda/m + o(m^{-1})$ . The interpretation becomes incorrect if that excess-risk result is treated as a total-error result. The two statements are

$$\mathcal{R}_m - \omega^2 = \frac{\Lambda}{m} + o(m^{-1}), \quad (46)$$

$$\mathcal{R}_m = \omega^2 + \frac{\Lambda}{m} + o(m^{-1}). \quad (47)$$

Only the first difference converges to zero. *The first rate is a proposal claim; the second equality is derived from that claim and (8).*

Thus Appendix A.2’s phrase “identifies everything” is correct only if “everything” means the shared parameters and the posterior rule. It is incorrect if it includes the realized vector  $(\theta_1, \dots, \theta_J)$ ,

the realized effect of an unvalidated cell, or the exact true ranking of cells. The model identifies posterior means, intervals, and ranking probabilities. It does not point-identify those latent realizations. Aggregate quantities may nevertheless be precise when many independent cell residuals average out, but that is a different asymptotic argument and does not imply precise recovery of each cell. *Derived from (39) and the distinction between cell-level and aggregate error.*

The proposal therefore contains a mismatch rather than an error in the displayed  $m^{-1}$  formula. Its formal result establishes convergence to the score-only oracle, while its phrases “recover the map” and “identifies everything” suggest convergence to the realized truth. If precise recovery of every unvalidated  $\theta_j$  is the intended deliverable, the current information structure cannot deliver it. A correct statement would say that the validation anchor consistently estimates the common calibration channel and produces a posterior distribution for each unvalidated cell, with limiting conditional variance  $\omega^2$ .

## 4 What can reduce the cell-level floor

### Notation

| Symbol                                 | Meaning                                                                                                                       |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| $j, m, J, R_j$                         | Cell index, validation count, total cells, and validation indicator.                                                          |
| $\theta_j, S_j, \hat{v}_j, s_j^2$      | True gain, broad score, direct RCT estimate, and its error variance.                                                          |
| $\tau^2, a, b, \rho, \psi_0, \omega^2$ | Cross-cell variance, score intercept and slope, score-truth correlation, true shared parameter, and current score-only floor. |
| $\Delta_m$                             | Shared-parameter component of held-out MSE.                                                                                   |
| $\ell = 1, \dots, n_j$                 | Cheap score observation and number of such observations in cell $j$ .                                                         |
| $S_{j\ell}, \bar{S}_j$                 | Conversation-level score and its cell average.                                                                                |
| $B_j, \sigma_B^2$                      | Systematic cell-level score error and its variance.                                                                           |
| $e_{j\ell}, \bar{e}_j, \sigma_e^2$     | Idiosyncratic score error, its cell average, and its variance.                                                                |
| $\omega_j^2(n_j)$                      | Posterior variance after averaging $n_j$ cheap score observations.                                                            |
| $Z_j, \zeta_j, q_j^2$                  | Additional cheap causal measurement for cell $j$ , its error, and its error variance.                                         |
| $X_j, \omega_X^2$                      | Additional cell characteristics and the remaining conditional variance after using them.                                      |
| $r_\star$                              | Desired score-only RMSE for an unvalidated cell.                                                                              |
| RMSE                                   | Root mean squared error.                                                                                                      |

For a validated cell in the primary calibration, (10) gives

$$\text{Var}(\theta_j | S_j, \hat{v}_j, \psi_0) = \frac{80.64(9)}{80.64 + 9} = 8.10, \quad \text{RMSE} = 2.85. \quad (48)$$

*Derived.* The direct measurement therefore changes cell-level precision materially. But with  $m = 40$  and  $J = 1,000$ , only 4 percent of cells receive it; 96 percent remain score-only under the basic design.

The project already has cheap data for every cell through  $S_j$ . The first question is therefore not whether another broad score exists. It is whether the error in the existing score averages away as the number of conversations in the cell grows.

To separate sampling error from systematic cell error, write

$$S_{j\ell} = a + b\theta_j + B_j + e_{j\ell}, \quad \bar{S}_j = a + b\theta_j + B_j + \bar{e}_j, \quad (49)$$

with mutually independent Gaussian components for this diagnostic model,

$$\text{Var}(B_j) = \sigma_B^2, \quad \text{Var}(e_{j\ell}) = \sigma_e^2. \quad (50)$$

Then

$$\omega_j^2(n_j) = \left[ \frac{1}{\tau^2} + \frac{b^2}{\sigma_B^2 + \sigma_e^2/n_j} \right]^{-1}. \quad (51)$$

*Derived under the diagnostic Gaussian model.*

If  $\sigma_B^2 = 0$ , then  $\omega_j^2(n_j) \rightarrow 0$ : enough cheap observations can identify  $\theta_j$  precisely. If  $\sigma_B^2 > 0$ , then

$$\omega_j^2(n_j) \rightarrow \left( \frac{1}{\tau^2} + \frac{b^2}{\sigma_B^2} \right)^{-1} > 0. \quad (52)$$

More conversations remove  $e_{j\ell}$  but not  $B_j$ . The immediate empirical task is therefore to estimate whether validation residual variance falls with cell interaction counts or instead plateaus. Repeated scoring and independent scoring methods can also separate transient variation from shared systematic error. *Design implication.*

Improving the score can solve the problem only if it is substantial in the real data. From  $\omega^2 = \tau^2(1 - \rho^2)$ , attaining a desired score-only RMSE  $r_\star$  requires

$$\rho \geq \sqrt{1 - \frac{r_\star^2}{\tau^2}}. \quad (53)$$

With  $\tau = 10$ , the requirements are:

| Desired $r_\star$   | Required $\rho$ |
|---------------------|-----------------|
| 5 percentage points | 0.866           |
| 3 percentage points | 0.954           |
| 2 percentage points | 0.980           |

*Derived.* A favorable correlation chosen only inside a simulation is not a fix; the observed score must actually attain that reliability.

If the existing score retains systematic error, four changes are possible.

1. *Improve the cell-specific predictors.* Add measured task complexity, baseline duration, verifiability, workflow, user composition, and other characteristics  $X_j$ . The new floor is

$$\omega_X^2 = \mathbb{E}[\text{Var}(\theta_j | S_j, X_j)]. \quad (54)$$

This helps only if the conditional residual variance actually falls on held-out validation cells. A flexible prediction method cannot remove a cell-specific residual that its inputs do not measure.

2. *Give important cells a cheap causal signal.* A broad platform randomization or randomized rollout could yield

$$Z_j = \theta_j + \zeta_j, \quad \text{Var}(\zeta_j) = q_j^2. \quad (55)$$

Combining it with the calibrated score gives

$$\text{Var}(\theta_j | S_j, Z_j) = \left( \frac{1}{\omega^2} + \frac{1}{q_j^2} \right)^{-1}. \quad (56)$$

This is an additional cell-specific causal measurement, not another copy of the existing  $S_j$ . When universal randomization is infeasible, scarce experimental observations can be allocated toward cells with the largest posterior uncertainty or greatest decision value. This improves those cells; it does not identify a cell that still receives no new information.

3. *Impose and test shared structure.* Occupation-by-task effects may have a low-dimensional or low-rank structure. Experiments in selected cells could then reveal components shared with an unvalidated cell. This replaces the exchangeable random-effects model with a stronger restriction. It is useful only to the extent that held-out-cell residuals from that structure are small; otherwise a new floor remains.
4. *Use credible quasi-experimental variation.* Staggered rollout, eligibility rules, capacity constraints, or randomized encouragement can provide cell-specific treatment variation without a separate enterprise RCT. The resulting estimate is causal only under the assumptions appropriate to that source of variation; ordinary observational prediction is not a substitute.
5. *Change the reported estimand or precision claim.* If neither cell-specific causal information nor a well-supported predictive structure is available, report posterior means with their full intervals, or report effects for broader occupation or task groups. Pooling can identify a group mean more precisely, but it does not identify the original individual  $\theta_j$ .

Changing the simulation parameters can illustrate when the problem is large or small. It is not a substantive fix unless the real data become more informative. Increasing  $m$ , improving the RCT precision in other cells, or changing the empirical-Bayes estimator reduces  $\Delta_m$ ; none of these changes eliminates  $\omega^2$  for a cell that still supplies only the same noisy  $S_j$ .

## A Derivation of the held-out-cell MSE

### Notation

| Symbol                                            | Meaning                                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| $j, m, R_j$                                       | Held-out cell, validation count, and validation indicator.                                            |
| $\theta_j, S_j, \eta_j, \hat{v}_j, \varepsilon_j$ | True gain, score, score error, validation estimate, and validation error.                             |
| $\mu, \tau^2, a, b, \sigma_S^2$                   | The five shared Gaussian channel parameters.                                                          |
| $\psi, \psi_0, \hat{\psi}_m$                      | Generic, true, and estimated shared parameter vectors, ordered as $(\mu, \tau^2, a, b, \sigma_S^2)$ . |
| $\mathcal{D}_m, h, m_j, \hat{\theta}_{j,m}$       | Validation data, conditional-mean rule, oracle score-only mean, and fitted prediction.                |
| $\mathcal{R}_m, \omega^2, \Lambda$                | Total held-out MSE, oracle floor, and coefficient on $m^{-1}$ .                                       |
| $\dot{h}(S_j; \psi_0)$                            | Derivative of $h$ with respect to $\psi$ .                                                            |
| $V_\psi$                                          | Asymptotic second-moment matrix of $\sqrt{m}(\hat{\psi}_m - \psi_0)$ .                                |
| $r_{j,m}$                                         | Remainder in the first-order Taylor expansion.                                                        |
| $D_S, x_j$                                        | Score variance $b^2\tau^2 + \sigma_S^2$ and centered score $S_j - a - b\mu$ .                         |
| $\lambda$                                         | Score reliability ratio $b^2\tau^2/(b^2\tau^2 + \sigma_S^2)$ .                                        |
| $s^2, \hat{\mu}_m$                                | Common validation-error variance and validation-mean estimator in the final simple case.              |

| Symbol      | Meaning                                     |
|-------------|---------------------------------------------|
| $o(m^{-1})$ | A term that is small relative to $m^{-1}$ . |

The derivation uses four conditions.

M1 The Gaussian model (1) is correct.

M2 The held-out cell is not used to estimate the shared parameter:

$$\mathcal{D}_m \perp\!\!\!\perp (\theta_j, S_j) \mid \psi_0. \quad (57)$$

M3 The shared-parameter estimator satisfies

$$\sqrt{m}(\hat{\psi}_m - \psi_0) \Rightarrow N(0, V_\psi), \quad m\mathbb{E}[(\hat{\psi}_m - \psi_0)(\hat{\psi}_m - \psi_0)^\top] \longrightarrow V_\psi. \quad (58)$$

M4 The function  $h$  is differentiable at  $\psi_0$ ,  $\mathbb{E}[r_{j,m}^2] = o(m^{-1})$ , and cross terms containing the remainder are  $o(m^{-1})$ .

M1 is the proposal's model. M2–M4 are additional assumptions. The proposal does not state this complete set. *Scope qualification.*

Add and subtract  $m_j = h(S_j; \psi_0)$ :

$$\hat{\theta}_{j,m} - \theta_j = (m_j - \theta_j) + (\hat{\theta}_{j,m} - m_j). \quad (59)$$

Squaring and taking expectations gives

$$\mathcal{R}_m = \mathbb{E}[(m_j - \theta_j)^2] + \mathbb{E}[(\hat{\theta}_{j,m} - m_j)^2] \quad (60)$$

$$+ 2\mathbb{E}[(m_j - \theta_j)(\hat{\theta}_{j,m} - m_j)]. \quad (61)$$

The cross term is zero:

$$\mathbb{E}[(m_j - \theta_j)(\hat{\theta}_{j,m} - m_j)] \quad (62)$$

$$= \mathbb{E}\left[(\hat{\theta}_{j,m} - m_j)\mathbb{E}[m_j - \theta_j \mid S_j, \mathcal{D}_m, \psi_0]\right] \quad (63)$$

$$= 0. \quad (64)$$

The first equality is iterated expectation. The last equality uses M2 and  $\mathbb{E}[\theta_j \mid S_j, \psi_0] = m_j$ . Also,

$$\mathbb{E}[(m_j - \theta_j)^2] = \mathbb{E}[\text{Var}(\theta_j \mid S_j, \psi_0)] \quad (65)$$

$$= \omega^2. \quad (66)$$

Substitution proves the exact decomposition

$$\mathcal{R}_m = \omega^2 + \mathbb{E}[\{h(S_j; \hat{\psi}_m) - h(S_j; \psi_0)\}^2]. \quad (67)$$

*Derived from M1–M2.*

By M3–M4,

$$h(S_j; \hat{\psi}_m) - h(S_j; \psi_0) = \dot{h}(S_j; \psi_0)^\top (\hat{\psi}_m - \psi_0) + r_{j,m}. \quad (68)$$

Using M2 to separate the held-out score from the validation estimator,

$$\mathbb{E}[\{h(S_j; \hat{\psi}_m) - h(S_j; \psi_0)\}^2] \quad (69)$$

$$= \frac{1}{m} \mathbb{E}[\dot{h}(S_j; \psi_0)^\top V_\psi \dot{h}(S_j; \psi_0)] + o(m^{-1}) \quad (70)$$

$$= \frac{\Lambda}{m} + o(m^{-1}), \quad (71)$$

where

$$\Lambda = \mathbb{E}[\dot{h}(S_j; \psi_0)^\top V_\psi \dot{h}(S_j; \psi_0)]. \quad (72)$$

Substitution into (67) proves

$$\mathcal{R}_m = \omega^2 + \frac{\Lambda}{m} + o(m^{-1}). \quad (73)$$

*Derived from M1–M4.*

For completeness, let

$$D_S = b^2 \tau^2 + \sigma_S^2, \quad x_j = S_j - a - b\mu. \quad (74)$$

Direct differentiation of (4) gives

$$\dot{h}(S_j; \psi_0) = \begin{pmatrix} \frac{\sigma_S^2}{D_S} \\ \frac{b\sigma_S^2 x_j}{D_S^2} \\ -\frac{b\tau^2}{D_S} \\ \frac{\tau^2(\sigma_S^2 - b^2 \tau^2)x_j}{D_S^2} - \frac{b\tau^2 \mu}{D_S} \\ -\frac{b\tau^2 x_j}{D_S^2} \end{pmatrix}. \quad (75)$$

The rows follow the order  $(\mu, \tau^2, a, b, \sigma_S^2)$ . Hence  $\Lambda$  is determined once the validation design determines  $V_\psi$ . *Derived.*

A simple case makes the  $m^{-1}$  term transparent. Suppose  $\mu$  is the only unknown shared parameter and

$$\hat{\mu}_m = \frac{1}{m} \sum_{j: R_j=1} \hat{v}_j, \quad \text{Var}(\varepsilon_j) = s^2, \quad (76)$$

with independent validation cells and errors. Then

$$\text{Var}(\hat{\mu}_m) = \frac{\tau^2 + s^2}{m}, \quad \frac{\partial h}{\partial \mu} = \frac{\sigma_S^2}{b^2 \tau^2 + \sigma_S^2} = 1 - \lambda. \quad (77)$$

The formula is exact in this case:

$$\mathcal{R}_m = \omega^2 + \frac{(1 - \lambda)^2 (\tau^2 + s^2)}{m}. \quad (78)$$

*Derived.*

## Conclusion

The broad score  $S_j$  is observed in every cell. The randomized anchor learns how that score relates to true gains, and the simulation confirms that its shared-parameter error falls approximately as  $1/m$ . Under the proposal calibration, that error is already small near  $m = 40$ ; almost all remaining held-out-cell MSE is  $\omega^2$ .

The existence of a positive floor comes from the current score-plus-random- anchor model. Its numerical dominance is parameter-dependent. When it dominates, an unvalidated  $\theta_j$  is not merely difficult to estimate: it is not consistently estimable from  $(S_j, \mathcal{D}_m)$ . A defensible fix must show that repeated cheap score data remove the score error, add cell-specific information, impose a validated structure that makes the residual small, or weaken the cell-level precision claim.

This establishes the proposal’s exact overreach. The randomized anchor can identify the shared channel and make the feasible predictor approach the oracle. It cannot, under the same information structure, make that oracle equal the realized effect in an unvalidated cell.